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ENGINEERING ROOT-CAUSE REPORT

CV-7 Brake Control Module — Regenerative/Friction Blend Transition Defect (BCM-FW 4.2.x)

Field	Detail
Report no.	ENG-RCA-2025-117
Subsystem	Brake Control Module (BCM); regenerative braking torque blending
Affected firmware	BCM-FW 4.2.0 – 4.2.3
Corrective build	BCM-FW 4.3.0
Date	12 October 2025
Prepared by	Caroline Stubbs (BCM Software Lead); approved by Sophie Lindqvist (CTO)
Classification	Safety-related defect

1. Summary

1.1 This report sets out the root cause of the transient braking under-application identified on the CV-7 and the corrective action taken. The defect arises from a boundary-condition error in the BCM torque-blending look-up table that, under a specific low-friction / high-brake-demand manoeuvre, causes regenerative torque to be reduced before friction-brake pressure has fully ramped, producing a short window (approximately 300–600 ms) of reduced total deceleration. Corrective firmware build 4.3.0 eliminates the defect and has passed the extended validation suite.

2. System description

2.1 The CV-7 BCM computes total commanded deceleration from the brake-pedal position and vehicle dynamics, and apportions it between regenerative (motor) braking and friction (hydraulic) braking. Apportionment uses a two-dimensional look-up table (the 'blend map') indexed by (i) estimated road-surface friction coefficient (μ), derived from wheel-slip and traction signals, and (ii) the rate of change of brake demand (jerk).

2.2 During normal operation the BCM blends smoothly between sources. As μ falls or demand rises rapidly, the BCM is designed to transfer braking from the motor to the friction brakes (which are more controllable near the limit of adhesion), ramping friction pressure up as regenerative torque is ramped down so that total force is preserved.

3. The defect

3.1 Observed behaviour

On a low-μ surface (μ ≤ 0.3, e.g. wet or icy) combined with a rapid increase in brake demand from light to heavy, telematics show regenerative torque being commanded down ahead of the friction-pressure ramp, so that for ~300–600 ms the sum of regenerative and friction braking is below the commanded value. Measured peak deceleration shortfall in track testing was up to 1.4 m/s², increasing stopping distance from 50 km/h on a wet surface by an estimated 2–4 m in the worst reproduced case.

3.2 Root cause

The blend map contains a boundary cell at the intersection of the lowest-μ band and the highest-jerk band. In build 4.2.0 the map was re-calibrated to improve energy recovery. The re-calibration set the regenerative-torque target for that boundary cell using the interpolation from the adjacent (higher-μ) cells, without applying the friction-pressure pre-charge offset that the lower-μ path requires. As a result the BCM, on entering that cell, commands the regenerative ramp-down on the higher-μ timing while the friction ramp follows the (slower) default schedule, opening the transient gap.

3.3 Why it was not detected earlier

The regression test suite in use at build 4.2.0 exercised low-μ manoeuvres and high-jerk manoeuvres separately, but not the combined low-μ / high-jerk manoeuvre that exposes the boundary cell. The defect therefore passed validation. The suite has been extended (see section 5).

4. Corrective action (build 4.3.0)

1. The boundary-cell regenerative-torque target has been corrected and the friction-pressure pre-charge offset is now applied across the full low-μ band.
2. A guard has been added so that, whenever μ is estimated below 0.35, friction-pressure pre-charge leads any regenerative ramp-down, guaranteeing total-force continuity irrespective of map values.
3. Telemetry has been added to log blend-transition force continuity for ongoing field monitoring.

5. Validation

Test	Result (4.2.3)	Result (4.3.0)
Combined low-μ / high-jerk, wet (μ≈0.25)	FAIL — 1.4 m/s² shortfall	PASS — <0.1 m/s² shortfall
Combined low-μ / high-jerk, icy (μ≈0.15)	FAIL — 1.1 m/s² shortfall	PASS — within tolerance
Standard FMVSS/UNECE brake performance	PASS	PASS
Energy-recovery efficiency (regression)	Baseline	−0.4% vs 4.2.3 (acceptable)
Extended regression suite (812 cases)	n/a	PASS (812/812)

6. Conclusion

6.1 The defect is a calibration boundary error, fully understood and reproduced, and is corrected by build 4.3.0 with no material loss of energy-recovery performance. Engineering supports the classification of the matter as a safety-related defect and the deployment of build 4.3.0 across the affected population of 2,140 vehicles by over-the-air update and workshop reflash.

Sophie Lindqvist

Chief Technology Officer, Calderwell Mobility Limited